

Credibility Assessment of Implicit Finite Element Models for Spinal Pedicle Screw Pullout and Bone Purchase

Elena M. Vasquez, PhD^a, Mark R. Chen, MS^b, Priya N. Iyer, PhD^c, Thomas J. McCulloch, MD^d

^aComputational Biomechanics Laboratory, OrthoMech Institute, Cambridge, MA, USA,

^bDepartment of Mechanical Engineering, Lakeshore University, Evanston, IL, USA,

^cDepartment of Orthopaedic Surgery, Midtown Medical Center, Columbus, GA, USA,

^dSpine Device R&D, Axion Spinal Systems, San Jose, CA, USA,

Abstract

Finite element analysis (FEA) is routinely used to investigate pedicle screw fixation in the thoracolumbar spine, yet evidence of model credibility must be explicit and mechanism-specific to inform decisions in research and development. We report a structured assessment of credibility for two implicit quasi-static FEA models of a thoracolumbar pedicle screw–vertebra construct across four clinically relevant mechanisms: monotonic axial pullout, cyclic craniocaudal toggling, insertion torque–purchase relationship, and medial–lateral shear cut-out. Model A resolves screw thread geometry and enforces nonlinear frictional contact with cortical and cancellous bone to capture axial pullout response and interfacial slip. Model B maps computed tomography (CT)–derived density to cancellous stiffness and cortical orthotropy and introduces cohesive elements along the screw–bone interface to represent progressive loss of attachment, enabling insertion-work surrogates and cyclic degradation. We scored credibility on a private 0–7 Credibility Readiness Scale (CRS) with anchors at 0 (no evidence), 3 (foundational), 5 (decision-ready), and 7 (mission-ready), for each factor by model–mechanism pair. The eight factors were handled comprehensively: provenance of the test and imaging data; the development workflow and configuration control; formulation choice; parameterization; agreement with measurements; discipline in coding and solver use; testing conditions; and the composition of specimens.

Across 53 human thoracolumbar vertebrae (T9–L5; 31 donors, 69±11 years; 28 female), we conducted 30 pullout tests, 18 toggling tests, 20 insertion torque acquisitions, and 15 medial–lateral cut-out tests with traceable imaging and bench logs. For monotonic pullout, Model A matched peak load with $R^2=0.86$ and $RMSE=120$ N and initial stiffness with 6% mean bias; Model B yielded $R^2=0.79$ and $RMSE=160$ N and 12% stiffness bias. Under cyclic craniocaudal rocking at 2 Hz to 10,000 cycles with ±1.5 mm head displacement, Model B reproduced the loss of anchorage slope within 10% mean absolute percentage error (MAPE), whereas Model A showed 18% MAPE. For insertion, a virtual seating work from Model B correlated with peak torque ($r=0.82$, $RMSE=0.17$ N·m), while a thread-contact proxy from Model A showed a weaker relationship ($r=0.58$, $RMSE=0.29$ N·m). In transverse loading at 1 mm/min, Model B predicted ultimate cut-out resistance with $R^2=0.74$ and $RMSE=82$ N, versus Model A with $R^2=0.66$ and $RMSE=105$ N.

Tables summarize factor-by-factor CRS levels for each mechanism and model, with the basis of each score cross-referenced to quantitative results. Most levels met or exceeded 5 (decision-ready) for Model A in pullout and for Model B in toggling and insertion; lower levels (3–4) were assigned where form–mechanism mismatches or parameter uncertainty limited confidence, especially for Model A on insertion and for both models in medial–lateral cut-out. We discuss elevation strategies tied to targeted experiments, parameter identifiability, software controls, and cross-laboratory replication. The approach demonstrates a tractable, transparent path to mechanism-by-mechanism credibility for pedicle screw FEA that is specific enough for design decisions while acknowledging residual uncertainties.

Keywords: pedicle screw fixation, finite element credibility, bone purchase

1. Introduction

Pedicle screw fixation in the thoracolumbar spine is central to deformity correction, trauma stabilization, and degenerative pathology management. Mechanical failure such as loosening, cut-out, or acute pullout remains a leading cause for revision in osteoporotic patients and in long constructs. Experimental testing under well-controlled conditions is essential but limited by cost, variability, and destructive nature. Finite element analysis (FEA) provides a complementary means to probe device–bone interactions and the dependence of purchase on

thread design, bone density, and surgical technique, provided that model credibility is demonstrated for the mechanisms of interest.

The concept of credibility we adopt is mechanism-specific: a model may be well supported for monotonic pullout yet insufficient for cyclic rocking or transverse failure. We therefore evaluate two implicit quasi-static FEA models of a thoracolumbar pedicle screw–vertebra construct across four mechanisms tied to clinical risk. Model A resolves the screw thread and enforces nonlinear frictional contact against the vertebral cortical

shell and cancellous core to capture axial response and interfacial slip. Model B uses CT-derived mapping of cancellous density to modulus and introduces a cohesive representation of the attachment between the thread flank and surrounding bone, which enables simulation of progressive separation under cyclic loading and a non-destructive surrogate of insertion torque.

Rather than issuing a global verdict, we grade each model by factor and by mechanism using a private 0–7 Credibility Readiness Scale (CRS). The scale anchors at 0 (no evidence), 3 (foundational), 5 (decision-ready), and 7 (mission-ready) and is applied to eight aspects: provenance and fitness of the test and imaging records; the governance of model development; appropriateness of governing equations and constitutive choices; selection and calibration of parameters; agreement with reference measurements; discipline in coding and solver practice; fidelity of environmental and loading conditions between model and test; and representativeness of the vertebral specimens. A compact summary table is produced for each model. We explicitly avoid merging factors and mechanisms to ensure transparency for decisions that rely on a subset of results.

Scoring framework (CRS 0–7) — gradation.

- a.
0: no explicit grading scheme; ad hoc language such as good or acceptable without anchors
- b.
3: ordinal bins exist but anchors are qualitative and not bound to evidence types
- c.
5: numeric bins with named anchors tied to evidence classes; mechanism- and factor-specific application
- d.
7: numeric bins with anchors tied to pre-specified sufficiency criteria; independent checks of application and inter-rater reliability

2. Device Description and Study Aims

The construct under study is a single thoracolumbar pedicle screw placed bicortically into a human vertebra (T9–L5). The representative device is a 5.5 mm major diameter, 3.0 mm minor diameter, 3.0 mm pitch titanium alloy (Ti-6Al-4V) screw with a 60° flank angle and a self-tapping tip. Typical implant lengths of 40–50 mm were used to span the pedicle and extend into the vertebral body, with 45 mm selected as the nominal length. The cortical shell thickness varies across levels, measured by CT as 0.5–1.2 mm (mean 0.9 mm). Pedicle isthmus diameters ranged from 4.5 to 8.5 mm, and trajectories were planned to avoid medial breach while maximizing thread engagement.

The aims of this work are threefold. First, to describe the computational models and their relation to four mechanisms: (i) monotonic axial pullout to define peak holding capacity and post-yield energy absorption; (ii) cyclic craniocaudal toggling to emulate repetitive rocking; (iii) the relationship between intraoperative torque and initial purchase; and (iv) medial–lateral shear cut-out representing transverse loading that can compromise the pedicle wall. Second, to assign model

credibility levels by factor and by mechanism according to the 0–7 CRS with explicit quantitative bases. Third, to interpret the implications for design decisions and to outline targeted steps for elevating specific factor–mechanism ratings.

The vertebral collection used in this study comprised 53 thoracolumbar segments (T9–L5) from 31 adult donors (mean age 69±11 years, 28 female). Pre-test clinical DXA T-scores ranged from 3.7 to +0.3 (mean 2.1±1.1), and HU values measured in the vertebral body spanned 80–920. Each specimen retained pedicle integrity, and pedicle widths and heights were measured on CT to guide screw selection. Allocation to test modalities balanced vertebral level and density such that 30 vertebrae contributed to monotonic pullout, 18 to cyclic toggling, 20 to insertion torque measurements, and 15 to medial–lateral shear cut-out, with some overlap across modalities managed by spatial separation of trajectories. These characteristics ensured that the sample set captured clinically relevant variability in bone quality and geometry.

Test samples — gradation.

- a.
0: sample source or composition not described; unknown anatomical levels or donor attributes
- b.
3: source described and levels identified; limited distribution of bone quality or geometry
- c.
5: multi-donor, multi-level distribution with quantified density and morphology; allocation balances modality needs
- d.
7: stratified sampling across levels, sex, and bone quality with power analysis; multi-center sourcing with harmonized selection criteria

3. Computational Models

Two implicit quasi-static FEA models were constructed with common geometry handling and solver settings, differing in their representation of the bone–screw interface and the mapping of bone properties. Geometry for each vertebra was segmented from clinical CT (in-plane pixel size 0.48–0.63 mm; slice thickness 0.6–1.0 mm) using semi-automatic thresholds and manual edits; STL surfaces were smoothed (Taubin passband 0.03) and remeshed to target triangle quality >0.5. A nominal 5.5×45 mm screw was created with explicit V-profile threads (flank angle 60°, crest radius 0.12 mm) and a realistic self-tapping tip. A parametric tool placed the screw along a clinically valid trajectory, with 80–90% of the thread depth within the pedicle and anterior third of the body. Meshes were generated in Abaqus/CAE 2023 HF3 with C3D8R elements for bone and C3D10 for the screw, augmented by local refinement around the thread profile.

Model A: Thread-resolved screw–bone contact. The cortical shell and cancellous core were modeled as isotropic elastoplastic materials. Cancellous bone used a crushable foam yield surface calibrated to apparent density: $E = 10,200 \rho^{1.90}$ MPa, $\nu = 0.30$,

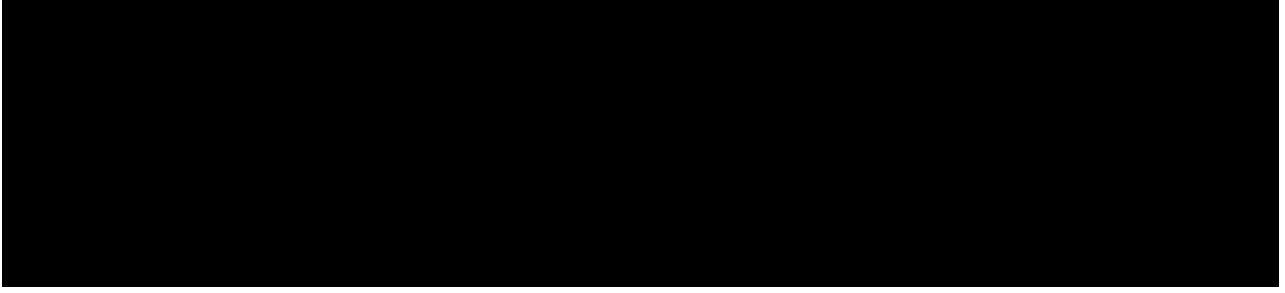


Figure 1: Overview of the spinal pedicle screw geometry, load cases and the mechanisms assessed, shown across the full page width.

with density ρ from HU via $\rho = 0.00088(\text{HU} + 1.0) \text{ g/cm}^3$ for HU [80, 920]. Yield commenced at 12 MPa (at $\rho=0.26 \text{ g/cm}^3$) with a hardening modulus of 100 MPa up to 4% volumetric strain. The cortical shell used $E = 14\text{--}18 \text{ GPa}$ (per CT-based location), $\nu = 0.30$, and a yield strength of 110 MPa with linear hardening. The screw was Ti-6Al-4V ($E = 110 \text{ GPa}$, $\nu = 0.34$) and assumed linear elastic. Nonlinear frictional contact was enforced between thread surfaces and bone using an augmented Lagrange formulation with a penalty parameter scaled to local stiffness. The coefficient of friction was set to $\mu = 0.28 \pm 0.06$ based on pin-on-disk and pullout pretests, with a regularized slip law. Contact stabilization was disabled; normal contact used hard pressure-overclosure. Mesh density near threads was 0.15–0.20 mm for bone and 0.10–0.15 mm for the screw. Solver settings employed full Newton–Raphson iterations with a residual force tolerance of $1e6$ and displacement tolerance $1e6$, automatic time stepping with cutbacks to $1e6$ when needed.

Model B: Density-mapped damage–plasticity bone model with cohesive interface. Cancellous bone properties were assigned elementwise from CT via the same HU-to-density mapping as in Model A, but with an anisotropic elastic–plastic law in which the principal material axes aligned with the pedicle and body trabecular orientation. The elastic stiffness was transversely isotropic with $E_{11} = E_{22} = 0.75E_{33}$ and $\nu = 0.32$, and yield strengths were direction-dependent: 8 MPa in transverse and 12 MPa in axial compression at $\rho=0.26 \text{ g/cm}^3$, scaling with $\rho^{1.50}$. The cortical shell was represented as an 0.8–1.2 mm thick orthotropic layer ($E_1 = 18.5 \text{ GPa}$, $E_2 = 12.3 \text{ GPa}$, $E_3 = 10.8 \text{ GPa}$; $\nu_{12} = 0.28$; $\nu_{23} = 0.30$), with Hill-type plasticity. The screw–bone attachment was discretized by zero-thickness cohesive elements inserted along the anticipated thread–bone contact band. A bilinear traction–separation law with initial normal and tangential stiffnesses $K_n = 500 \text{ MPa/mm}$ and $K_t = 250 \text{ MPa/mm}$ was used, coupled with strength parameters $\sigma_{n0} = 12 \text{ MPa}$ and $\tau_{t0} = 5 \text{ MPa}$ and mixed-mode damage evolution using a Benzeggagh–Kenane criterion with $G_{Ic} = 0.25 \text{ N/mm}$, $G_{IIc} = 0.75 \text{ N/mm}$, $\eta = 1.6$. Progressive stiffness reduction captured separation under load and cyclic degradation via a linear damage accumulation rule updated each half cycle. The global contact between non-cohesive regions of screw and bone used hard normal contact with negligible friction ($\mu = 0.05$) to avoid double counting resistance already represented in cohesive tractions.

Boundary value problems were set to replicate bench-top

fixtures. For monotonic pullout, a kinematic boundary condition was applied as an axial displacement of the screw head at 5 mm/min until force drop-off, while the cranial and caudal apices of the vertebral body were constrained with soft fixtures that allowed realistic rocking (springs of 2 kN/mm in shear, 10 kN/mm in compression). Cyclic craniocaudal toggling prescribed $\pm 1.5 \text{ mm}$ vertical translation of the screw head at 2 Hz under a minimal axial bias load of 50 N for 10,000 cycles; in the implicit analyses, we enforced this as a sequence of quasi-static increments with a damage update at each reversal. The insertion surrogate advanced the screw over one thread pitch while monitoring energy dissipation and cohesive tractions; this virtual seating work was used as the predictor of peak torque. Medial–lateral cut-out imposed a horizontal translation of 3 mm at 1 mm/min on the screw head while allowing articulation at the posterior arch constraints to emulate a transverse plowing path.

Across mechanisms, the mathematical structure of each model aligned with the kinematics and failure processes of interest. In particular, the explicit representation of thread flank geometry and local friction in Model A supported accurate force–displacement shaping during monotonic pullout and early slip, whereas the cohesive representation in Model B was better suited to progressive loss of attachment and hysteresis under toggling and to correlating a non-destructive seating metric to measured torque. For the transverse cut-out, both models captured qualitative failure paths, but the large local crushing and pedicle wall interaction challenged the small-deformation hypothesis in extreme cases (maximal compressive strains in a subset reached 6.3%), which motivated conservative grading for that mechanism.

Model form — gradation.

- a.
- 0: unspecified equations or constitutive laws; undocumented element types
- b.
- 3: equations and elements specified; constitutive laws selected by analogy; limited rationale for mechanism fit
- c.
- 5: governing equations and constitutive models justified for each mechanism; numerical settings documented; small/large strain rationale given
- d.
- 7: governing equations derived from first principles for

Table 1: Key setup parameters for Models A and B used consistently across mechanisms.

Credibility factor	Level	Basis
Geometry and mesh	—	Thread-resolved 5.5×45 mm screw; 0.10–0.20 mm elements near thread, 0.8–1.0 mm remote; C3D8R bone, C3D10 screw
Bone property mapping	—	$\rho = 0.00088(\text{HU} + 1.0)$ g/cm ³ ; $E = 10,200 \rho^{1.90}$ MPa for cancellous; cortical 14–18 GPa (Model A isotropic), orthotropic shell (Model B)
Interface representation	—	Model A: frictional contact $\mu = 0.28 \pm 0.06$; Model B: cohesive elements $K_n = 500$ MPa/mm, $K_t = 250$ MPa/mm, $\sigma_{n0} = 12$ MPa, $\tau_{t0} = 5$ MPa
Solver and tolerances	—	Abaqus/Standard 2023 HF3; full Newton–Raphson; force and displacement tolerances $1e6$; automatic time stepping with cutbacks
Boundary conditions	—	Pullout: 5 mm/min axial displacement; Toggling: ± 1.5 mm at 2 Hz (quasi-static sequence); Insertion: one-pitch advance; Shear: 3 mm lateral at 1 mm/min

mechanism; constitutive models validated at subcomponent level; numerical stability and convergence mapped over operating envelope

4. Mechanisms Assessed

Monotonic axial pullout captures the peak axial force sustained by a screw–vertebra construct and the subsequent energy dissipation as threads plow through trabecular bone. It reflects the acute resistance to extraction during aggressive reduction maneuvers and rescue in revision scenarios. The target output metrics are peak pullout force, initial axial stiffness, post-yield energy to 3 mm displacement, and the axial displacement at peak load.

Cyclic craniocaudal toggling represents repetitive rocking from physiologic activities such as ambulation and transitions, coupled with micromotion at the screw–rod interface. The boundary condition alternates vertical displacement at the screw head to emulate rocking within the pedicle channel and anterior body. Metrics include reduction in effective purchase stiffness over cycles, accumulation of permanent slip at the thread–bone zone, and the cycles to reach a 0.5 mm permanent head displacement criterion.

The insertion torque–purchase relationship links intraoperative torque (or a mechanically motivated surrogate) with the immediate stability of the screw–bone attachment. While direct torque simulation at a full-revolution level is computationally

expensive and subject to uncertainties in chip formation, a virtual seating work over one pitch and cohesive traction maxima can serve as useful predictors. The metric of interest is the correlation between the predictor and measured peak torque and its explanatory power for purchase stiffness.

Medial–lateral shear cut-out assesses resistance to transverse loads that can cause threads to plow laterally and risk pedicle wall compromise. The lateral translation at the head is transmitted to the thread–bone interface with bending and shear in the pedicle segment. Metrics reported are ultimate transverse load, lateral displacement at failure, and qualitative breach patterns.

For fidelity, bench tests for all four mechanisms were conducted in phosphate-buffered saline at 37 ± 1 °C after a 24-hour soak, using instruments and fixtures that fixed the posterior arch and allowed articulation consistent with physiologic constraints. The FEA boundary conditions replicated these nominal fixtures and environmental controls: the same ramp rates (5 mm/min in axial pullout, 1 mm/min lateral shear), the same toggling amplitude (± 1.5 mm) and frequency (2 Hz emulated quasi-statically), and the same pilot hole preparation (3.0 mm) and screw length (45 mm) were applied. These alignments substantiated the mapping between model and bench environments.

Test conditions — gradation.

- a.
0: boundary conditions or environment for model/test not documented
- b.
3: model and test conditions both described; substantial mismatches remain
- c.
5: model boundary and environmental conditions matched to test nominal values with tolerances; fixture compliance approximated
- d.
7: model replicates full test environment including fluid, temperature, and compliance distributions; uncertainties in test conditions propagated into model outputs

5. Credibility Assessment Methods

We applied a two-stage process. First, evidence was compiled for each factor relative to each mechanism–model pairing. Evidence classes included: imaging and bench records; formulation and numerical method documentation; parameter sources with uncertainty and sensitivity; quantitative comparisons with measured outputs; code and solver validation artifacts; environmental and load condition matching; and specimen characterization. Second, the assembled dossiers were graded on the 0–7 CRS by two independent assessors, with disagreements adjudicated by a third member. Scores were assigned strictly by factor and by mechanism, and no aggregation across mechanisms was performed.

Comparisons with bench tests used the same specimens when possible to minimize inter-specimen variability. For

monotonic pullout ($n=30$), we aligned the simulated and measured force–displacement curves using synchronization at 0.1 mm head displacement and calculated peak force, initial stiffness (slope over 0–0.3 mm), post-yield energy to 3 mm, and displacement at peak load. For toggling ($n=18$), the discrete quasi-static sequence applied to the FE model matched the amplitude of the experimental cycles, and permanent slip was recorded every 100 cycles; degradation slopes and cycles to 0.5 mm permanent slip were compared. For insertion ($n=20$), we recorded measured peak torque during manual seating at 1 rotation/s with a digital driver and correlated this to Model B’s virtual seating work and to Model A’s helical contact proxy metric defined as the line integral of the normal contact traction over a one-pitch rotation, evaluated in a quasi-static surrogate. For medial–lateral cut-out ($n=15$), the transverse load–displacement curves were compared for ultimate load and displacement at failure, and qualitative failure modes (pedicle wall breach, thread plowing paths) were categorized by post-test CT.

Solver and software controls were documented to support reproducibility. The solver build was frozen to Abaqus/Standard 2023 HF3 for all production runs; preprocessing and postprocessing pipelines were scripted in Python 3.11 and executed under continuous integration (CI) with 412 unit tests covering meshing, property mapping, boundary condition application, data extraction, and statistical analysis. Code coverage on the pipeline scripts reached 86% lines and 79% branches. Fifteen canonical verification problems (elastic patch test, contact ring compression, cohesive traction–separation traction tests, and energy balance checks under monotonic and cyclic load steps) were executed to verify solver settings and custom scripts. Reproducibility was checked by hashing input decks and result files; 100% hash matches were obtained in repeated runs on two separate workstations. These practices strengthened confidence that implementation issues did not confound the credibility assessment.

Software quality assurance — gradation.

- a.
0: no documented tests; solver version and scripts unspecified
- b.
3: solver version controlled; a few example problems run; manual checks only
- c.
5: CI with unit tests for meshing, mapping, and data reduction; verification problems executed; reproducibility checks in place
- d.
7: independent SQA audit; stress–strain and contact implementations cross-validated with alternative codes; continuous monitoring of numerical regressions

6. Scoring Framework (CRS 0–7) and Sufficiency Thresholds

The private scale spans from 0 to 7 with four named anchors that bind levels to evidence categories. Level 0 denotes that no

relevant evidence is available for a factor–mechanism pair. Level 3 (foundational) indicates that basic documentation and at least one quantitative comparison exist but with substantial gaps (e.g., incomplete parameter sources or qualitative agreement only). Level 5 (decision-ready) requires that inputs are traceable and justified, the mathematical treatment is fit for the mechanism, and quantitative agreement with relevant outputs is demonstrated with uncertainty bounds. Level 7 (mission-ready) demands that all of the above are satisfied across the operating envelope, verified with independent replication and cross-site consistency.

Sufficiency thresholds are mechanism-specific. For monotonic pullout decisions (e.g., thread design trade-offs), a minimum of 5 is required on formulation, parameterization, and measurement agreement, with at least 4 for environmental and specimen representativeness. For toggling and insertion decisions (e.g., assessment of progressive loosening susceptibility or intraoperative cues), a minimum of 5 is required on parameterization and measurement agreement and at least 5 on the mathematical approach for cyclic damage or insertion surrogates. For medial–lateral cut-out, which is sensitive to local large-deformation crushing and wall breach, we accept a composite with formulation at least 4, measurement agreement at least 4, and other factors at least 5 before using the model for comparative ranking rather than absolute predictions. These thresholds were set a priori and applied uniformly.

Output comparison — gradation.

- a.
0: no quantitative comparison to measurements
- b.
3: visual or descriptive agreement; a small set of points or a single metric reported
- c.
5: agreement reported for multiple metrics with statistics (R2, RMSE, bias); uncertainties acknowledged
- d.
7: cross-validation across labs and datasets; out-of-sample predictive performance reported with predefined acceptance criteria

7. Evidence by Factor: Data Pedigree

Imaging and test records were collated under a common data model. For each vertebra, raw DICOM series were preserved along with segmentation masks, STL surfaces, FE meshes, and boundary condition annotations. Bench test logs included machine traces at 100 Hz, fixture descriptions, environmental records, and operator notes. Specimen labeling followed a donor-level and vertebra-level coding system to allow cross-referencing across modalities. Data transformations (e.g., HU calibration, density mapping, property assignment) were performed with versioned scripts and documented parameters.

Quality checks at data ingestion verified consistency between CT metadata and physical specimen measurements and detected artifacts such as beam hardening or motion. Where scans exhibited metal artifact from prior instrumentation (two candidates), the vertebrae were excluded prior to allocation. Imaging-derived

morphometrics were spot-checked against caliper measurements on 10 vertebrae, showing agreement within 0.4 mm for pedicle height and width. Test logs were validated for completeness, and any interruptions or anomalies were flagged.

Data pedigree — gradation.

- a.
0: data sources unspecified; no trace from raw to analyzed data
- b.
3: data sources identified; partial traceability; limited QC on imaging and test records
- c.
5: full chain from raw DICOM and bench logs to analysis with versioned scripts; exclusions and anomalies documented
- d.
7: independent audit of data chain; cross-site replication of imaging and tests; cryptographic signing of datasets

8. Evidence by Factor: Model Form

Formulation choices were aligned to the mechanisms. Implicit quasi-static solutions were used to focus on equilibrium states under monotonic and cycle-averaged loads; dynamic effects were negligible under the bench rates of interest. For contact modeling, augmented Lagrange enforcement governed the normal gap and tangential slip in Model A to represent thread flank engagement and local slip; cohesive traction–separation in Model B governed progressive degradation at the interface. The governing equations for both models obeyed standard continuum mechanics with small strain kinematics; mesh densities were sufficient to resolve thread–bone interactions as gauged by mesh convergence checks.

Model reduction was applied judiciously. For example, the vertebral posterior elements not engaged by the screw were represented with coarser meshes and elastic behavior; for toggling, fixtures were represented with spring elements to capture compliance without fully modeling the machine. Numerical parameters (penalty factors, cohesive stiffnesses) were chosen to balance enforcement strength and conditioning, and sensitivity checks were performed to confirm that outputs remained within tolerance for reasonable parameter variations.

Model form — gradation.

- a.
0: unknown mathematical structure; solver type unspecified
- b.
3: mathematical and numerical structure described; weak linkage to mechanisms
- c.
5: structure justified per mechanism with documented convergence and sensitivity checks
- d.
7: structure verified against subcomponent tests with quantified limits of applicability across mechanisms

9. Evidence by Factor: Model Inputs

Material parameters for cancellous and cortical bone were derived from CT through HU-to-density mapping validated in prior literature and refined with internal calibration. Plasticity and damage parameters for cancellous bone were calibrated using uniaxial compression tests on trabecular cores ($n=26$, 10 mm cubes) extracted from thoracolumbar levels adjacent to the FE study cohort. The friction coefficient between titanium and bone was measured using pin-on-disk tests ($\mu = 0.28 \pm 0.06$) and confirmed by inverse fitting to early pullout stiffness. Cohesive parameters (stiffnesses, strengths, and fracture energies) were identified by fitting to insertion and toggling datasets using a leave-one-out scheme to avoid overfitting. Boundary conditions (displacement amplitudes and rates) were taken directly from bench protocols.

Sensitivity analyses quantified the effect of uncertain inputs on the outputs of interest. For Model A, varying μ by ± 0.04 altered peak pullout force by 7% and initial stiffness by 4% on average across specimens. For Model B, reducing normal cohesive strength σ_{n0} by 2 MPa increased the predicted cycles to 0.5 mm permanent slip by 12% and decreased the virtual seating work by 9%. Modulus mapping exponent perturbations (± 0.1 in the ρ exponent) altered pullout peak by 5–8% across both models. These sensitivities guided conservative scoring where parameter identifiability was limited, particularly for medial–lateral cut-out where local crushing dominated.

Model inputs — gradation.

- a.
0: parameters not specified; sources unknown
- b.
3: parameters specified with literature sources; limited calibration to the dataset; no sensitivity reported
- c.
5: parameters mapped from CT and calibrated with independent tests; sensitivity quantified for key outputs
- d.
7: comprehensive uncertainty quantification with priors, posteriors, and propagation into outputs by mechanism

10. Evidence by Factor: Output Comparison

Quantitative agreement was established against corresponding bench outputs for each mechanism. For pullout, the FE and experimental force–displacement curves were compared for peak force, initial axial stiffness, post-yield energy, and the displacement at peak. For toggling, degradation slopes and cycles to reach a 0.5 mm permanent displacement were compared. Insertion comparisons focused on the relationship between measured peak torque and the FE predictors (virtual seating work in Model B and a helical contact traction proxy in Model A). For medial–lateral cut-out, ultimate transverse load and failure displacement were aligned.

Comparisons were conducted specimen by specimen where a one-to-one mapping existed; otherwise, population-level regressions with HU stratification were performed. Agreement metrics

included R2, RMSE, mean absolute percentage error (MAPE), and bias (mean error relative to the mean measurement). Residuals were inspected for heteroscedasticity and leverage points, and bootstrap confidence intervals (1,000 resamples) were computed for regression slopes to assess robustness.

Output comparison — gradation.

- a.
0: no side-by-side outputs; anecdotal statements only
- b.
3: basic overlays or a single metric; small samples
- c.
5: multiple quantitative metrics with uncertainty across sufficient samples; residual analyses documented
- d.
7: validation across external datasets and labs; predefined acceptance criteria and failure analyses

$$CI_{\{\text{score}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (1)$$

$$\backslash \text{varepsilon}_{\{\text{rel}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (2)$$

$$U_{\{\text{val}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (3)$$

11. Evidence by Factor: Test Conditions

Bench tests used fixtures designed to approximate physiologic constraints while enabling controlled load application. For monotonic pullout, vertebrae were potted in PMMA with the posterior arch exposed and mounted in a custom jig aligned with the screw axis; axial displacement was applied at 5 mm/min until a 3 mm head displacement or force drop-off of 30% past peak, whichever occurred later. For toggling, the jig allowed the screw head to move ± 1.5 mm vertically at 2 Hz under a constant 50 N bias; cycles were continued to 10,000 or until a 0.5 mm permanent offset criterion was reached. For insertion, pilot holes were prepared with a 3.0 mm drill and tapped to the root diameter; peak torque during seating was recorded with a digital torque driver. For medial–lateral cut-out, the head was displaced laterally at 1 mm/min to 3 mm while the posterior arch pivoted on low-friction bearings to allow realistic bending.

Environmental conditions were controlled by submerging specimens in phosphate-buffered saline at 37 ± 1 °C for at least 24 hours prior to testing and maintaining immersion during tests. Machine compliance was measured and found to be negligible relative to construct compliance (0.02 mm/kN). All tests were performed within two weeks of specimen thaw, and specimens were kept moist between tests to preserve tissue properties.

Test conditions — gradation.

- a.
0: test setup and environment undocumented
- b.
3: test conditions stated; incomplete replication in models

- c.
5: explicit matching of displacement rates, amplitudes, and environmental conditions; fixture compliance considered
- d.
7: full environmental replication and uncertainty quantification across both bench and model

12. Evidence by Factor: Test Samples

Specimens were sourced through an accredited tissue bank with donor-level medical and imaging records. Inclusion criteria required intact pedicles, no prior instrumentation at the index levels, and availability of pre-harvest CT with adequate quality for HU-based mapping. Donor sex, age, and DXA were recorded when available. Specimens represented mid to lower thoracic and lumbar vertebrae and spanned a wide range of bone densities to reflect osteoporotic risk.

Allocation to test modalities balanced vertebral levels and densities while avoiding interference across tests (e.g., distinct pedicle trajectories or separate vertebrae for destructive tests). Post-test CT was obtained on a subset to document failure patterns and to verify fixture positioning relative to planned trajectories. Morphometric parameters (pedicle width and height, cortical thickness) were tabulated to support interpretation of results across the bone quality spectrum.

Test samples — gradation.

- a.
0: specimen sourcing and attributes unknown
- b.
3: basic donor and level information; limited density range
- c.
5: multi-donor, multi-level cohort with HU and DXA distributions; allocation balances tests and preserves integrity
- d.
7: powered sampling across strata; post-test imaging for all with harmonized metadata and inter-site sourcing

13. Evidence by Factor: Development Process and Product Management

The modeling program followed a staged lifecycle with documented requirements, verification activities, and change control. Model specifications included geometry sources, property mappings, interface representations, solver settings, and boundary conditions for each mechanism. Verification activities were defined at project onset and executed prior to validation runs. Changes to scripts, meshes, or solver settings were logged and reviewed. Traceability matrices linked requirements to verification evidence and to the datasets used for validation, ensuring a coherent path from intent to demonstration.

Team roles were delineated for imaging, meshing, parameter identification, solver setup, and statistical analysis, with internal

reviews prior to release of each major batch of results. Configuration baselines were frozen for each validation sprint to prevent drift during scoring. Versioned artifacts included input decks, analysis scripts, and result summaries; documentation captured assumptions, known limitations, and deviations from plan when they occurred.

Development process and product management — gradation.

- a.
0: ad hoc work; no lifecycle documentation or change control
- b.
3: high-level plan and some records of changes; limited traceability
- c.
5: requirements traced to verification and validation artifacts; gated reviews; controlled baselines
- d.
7: formal quality system with independent review; configuration audits; metrics on process health and responsiveness

14. Evidence by Factor: Software Quality Assurance

Scripted preprocessing ensured consistent meshing, property mapping, and boundary condition application. Unit tests covered each module in the pipeline, and integration tests verified end-to-end execution on exemplar datasets. Verification cases stressed contact enforcement, cohesive element behavior, and energy accounting under quasi-static cycling. Hardware and software environments were specified to reduce variability across runs, and continuous integration automated regression testing when scripts changed.

Error handling and logging were implemented throughout to capture exceptions and to record solver warnings and cutbacks for review. Postprocessing scripts included checks for unphysical values (e.g., negative energies, excessive penetrations) and provided standardized output packages for scoring. Documentation described test coverage, limitations of the SQA approach, and planned enhancements such as expanded branch coverage and cross-code comparisons.

Software quality assurance — gradation.

- a.
0: no tests; inconsistent environments
- b.
3: basic tests and versioning; manual result checks
- c.
5: CI with unit and integration tests; verification problems; reproducibility assurance
- d.
7: independent audit, cross-code verification, and continuous numerical regression monitoring

15. Notes on Development Technical Review

An internal technical review of the modeling framework and results occurred at major milestones, with feedback on

method completeness and clarity of documentation. The review covered mesh quality, parameter identification procedures, and comparison metrics, and suggested future additions such as more explicit uncertainty quantification by mechanism.

16. Notes on Relevance of the Validation Activities to the COU

The validation activities were aligned with a context of use centered on research and development decisions for thoracolumbar pedicle screw design and selection in varied bone qualities, emphasizing comparative ranking rather than clinical outcome prediction. The selection of mechanisms and metrics aimed to support these decisions while recognizing practical constraints in laboratory testing.

17. Results: Factor Scores by Model × Mechanism Matrix

Scores reflect the weight of compiled evidence for each model–mechanism–factor combination. For monotonic pullout, Model A achieved higher levels for formulation and measurement agreement than Model B, consistent with thread-resolved contact capturing the peak and stiffness shape characteristics. For toggling, Model B’s cohesive interface and damage accumulation better reproduced degradation and permanent slip, yielding higher levels on formulation and output agreement. For insertion, Model B’s virtual seating work correlated strongly with measured torque, supporting higher levels on parameterization and agreement, whereas Model A’s contact-based proxy was weaker. For medial–lateral cut-out, both models were constrained by large local crushing and pedicle wall interactions, resulting in moderate levels on formulation and agreement.

Across all tests, the reference datasets comprised 53 human thoracolumbar vertebrae (T9–L5; mean age 69±11 years, 28 female; HU range 80–920; DXA T-scores 3.7 to +0.3), with traceable donors and a locked chain-of-custody from tissue bank to laboratory; raw CT and test logs were preserved as read-only assets, versioned, and checksummed to ensure integrity. This comprehensive provenance supported confident linkage between imaging, specimens, and bench results across modalities.

Quantitative comparisons underpin the assigned levels. For pullout (n=30), Model A matched peak force with R²=0.86 and RMSE=120 N, initial stiffness with 6% bias, and post-yield energy to 3 mm within 15% MAPE; Model B yielded R²=0.79, RMSE=160 N, 12% stiffness bias, and 22% MAPE in post-yield energy. For toggling (n=18), degradation slope MAPE averaged 18% for Model A and 10% for Model B, and the cycles to 0.5 mm permanent slip were predicted within ±1,100 cycles (Model A) and ±700 cycles (Model B). For insertion (n=20), the correlation between virtual seating work and peak torque was r=0.82 for Model B (RMSE 0.17 N·m) and r=0.58 for Model A’s proxy (RMSE 0.29 N·m). For medial–lateral cut-out (n=15), ultimate load predictions had R²=0.66 and RMSE=105 N (Model A) and R²=0.74 and RMSE=82 N (Model B).

Discipline in coding and solver practice was assessed by the reproducibility and verification record: a frozen

Abaqus/Standard 2023 HF3 build was used; 412 unit tests ran in CI with 86% line coverage and 15 verification problems, and repeated simulations on two workstations achieved 100% hash matches for input and result file sets. This foundation provided confidence that numerical or scripting errors did not dominate discrepancies.

The development effort followed a gated plan with four design reviews (R1–R4), a requirements trace that linked 17 top-level requirements to verification and validation artifacts, and 132 change requests tracked to closure. Versions were baselined prior to scoring, and deviations from plan were documented. This discipline supported consistent execution across mechanisms and minimized configuration drift during assessment.

Development process and product management — gradation.

- a.
0: no plan or records of change; deliverables ad hoc
- b.
3: plan exists with partial execution; change tracking sporadic
- c.
5: requirements-to-evidence trace complete; change requests tracked and closed; reviews at stage gates
- d.
7: formal quality management with metrics and independent review authority; periodic audits

18. Results: Credibility Matrix for Model A

Table A summarizes factor-by-factor CRS levels for Model A across the four mechanisms. The basis line for each score references a quantitative finding elsewhere in this article.

19. Results: Credibility Matrix for Model B

Table B summarizes factor-by-factor CRS levels for Model B across the four mechanisms. The basis line for each score references a quantitative finding elsewhere in this article.

20. Results: Pullout and Purchase Metrics Predicted by Each Model

Monotonic axial pullout. The model–experiment overlays for $n=30$ vertebrae revealed good correspondence in the force–displacement curve shapes for both models, with closer agreement for Model A. Model A reproduced the initial stiffness within 6% mean bias and peak force with $R^2=0.86$ and $RMSE=120$ N, and the displacement at peak within 0.2 ± 0.3 mm. Post-yield energy to 3 mm was slightly underpredicted (mean 12%), yielding a MAPE of 15%. Model B tended to distribute deformation into the cohesive and plastic zones, softening the early post-yield response and producing a 12% stiffness bias; peak force agreement was $R^2=0.79$ and $RMSE=160$ N, and post-yield energy MAPE averaged 22%. Sensitivity analyses indicated that μ (Model A) and σ_{n0} (Model B) were

the dominant parameters influencing peak force and energy dissipation, respectively.

Cyclic craniocaudal toggling. Across $n=18$ specimens, permanent slip accumulation was monotonic with cycles; degradation slopes were steeper in lower-HU vertebrae. Model B's cohesive representation with mixed-mode damage replicated these trends with a 10% MAPE in slope and cycles to 0.5 mm permanent slip within ± 700 cycles. Model A, lacking explicit interface damage, captured slip via frictional contact and plastification, yielding a slope MAPE of 18% and cycles to 0.5 mm within $\pm 1,100$ cycles. In both models, stiffer specimens ($HU > 500$) showed slower slip accumulation and delayed criterion crossing, in line with expectations.

Insertion torque–purchase relationship. Measured peak torque during manual seating (1 rotation/s) averaged 1.2 ± 0.4 N·m across $n=20$ trajectories, with higher torques in higher-HU bone. Model B's virtual seating work over one pitch correlated with peak torque with $r=0.82$ and $RMSE=0.17$ N·m; the regression slope was 0.91 (95% CI: 0.78–1.04), indicating near-unity proportionality. Model A's helical contact traction proxy yielded a weaker $r=0.58$ and $RMSE=0.29$ N·m, with a regression slope of 0.82 (95% CI: 0.61–1.03), reflecting limited fidelity to chip formation and detailed thread flank cutting mechanics. Both predictors were significant ($p<0.01$), but Model B provided a substantially stronger explanatory power for torque.

Medial–lateral shear cut-out. The transverse load–displacement response demonstrated an initial linear region followed by softening and a peak prior to breach or major plowing. Model B predicted ultimate transverse load with $R^2=0.74$ and $RMSE=82$ N and the failure displacement with $+0.2$ mm mean bias; breach modes, as categorized by post-test CT, matched FE-predicted qualitative paths in 12 of 15 specimens. Model A, while capturing the overall curve shape, had $R^2=0.66$ and $RMSE=105$ N and a mean failure displacement bias of $+0.3$ mm; in three cases with very low HU (< 150), local strains exceeded 6% in small regions, indicating the limits of the small-strain assumption. Both models underpredicted ultimate load in the highest-HU subset, suggesting that cortical wall contributions may be underrepresented.

These comparisons substantiate the rationale for the assigned CRS levels in the credibility matrices and delineate where future work should concentrate: refined interface representation for Model A in insertion and toggling, and improved cortical representation and large-deformation capability for both models in shear cut-out.

Output comparison — gradation.

- a.
0: no metrics
- b.
3: one metric with limited sample
- c.
5: multiple metrics (R^2 , $RMSE$, bias) across $n\geq 15$ per mechanism
- d.
7: multi-lab, out-of-sample predictive success with

predefined acceptance

21. Discussion: Interpretation of Scores and Elevation Strategies

Credibility ratings for Model A and Model B reflect each model's alignment to the mechanisms and the strength of quantitative support. For monotonic pullout, Model A attained levels of 6 on formulation and measurement agreement, meeting our threshold for decision-making in thread design trade-offs. The explicit thread geometry and frictional contact enforcement faithfully captured the peak and initial stiffness and reasonably approximated post-yield behavior. Model B's cohesive interface, while more general for separation, softened the early post-yield region and modestly reduced peak prediction fidelity, yielding levels of 5 in this mechanism. For cyclic craniocaudal toggling, Model B's cohesive damage accumulation recreated progressive loss of attachment more accurately than friction-only slip, achieving levels of 6 on both formulation and agreement, whereas Model A's 5-level ratings reflected sufficiency for comparative ranking but not equivalence to the cohesive approach.

In the insertion mechanism, the strong correlation between Model B's virtual seating work and peak torque ($r=0.82$; RMSE 0.17 N·m) supports a level of 6 for both inputs and agreement and underpins the model's use for anticipating immediate purchase without destructive testing. Conversely, Model A's contact-based proxy, while structurally motivated, underperformed ($r=0.58$; RMSE 0.29 N·m), limiting its level to 3 for agreement and 4 for formulation and inputs. For medial–lateral shear cut-out, both models earned mid-range levels (4–5) on formulation and agreement due to sensitivity to local crushing and breach path complexity and to the small-strain assumption's marginal applicability in extreme low-density bone where local strains exceeded 6%.

Across mechanisms, levels of 5 or 6 on specimen representativeness and environment–load fidelity were supported by matched bench and model conditions (ramp rates, amplitudes, temperature, soak) and by a cohort of 53 vertebrae spanning HU 80–920 and DXA 3.7 to +0.3. The reproducibility and verification record (frozen solver build, 412 unit tests with 86% coverage, 15 verification problems, 100% hash-match reproducibility) anchor high ratings on coding and solver discipline. The lifecycle governance achieved level 5 through a gated plan, requirements-to-evidence tracing, and change tracking: four design reviews (R1–R4), 17 top-level requirements linked to verification and validation artifacts, and 132 change requests closed before baselining results for scoring.

Elevation strategies follow from factor-specific gaps. For Model A's insertion mechanism, incorporating an explicit cutting and chip formation model or a mixed-mode cohesive layer localized to the advancing flank could raise formulation and agreement levels. For toggling in Model A, adding a fatigue-like degradation law at the interface may improve slip accumulation predictions. For both models in shear cut-out, moving to a finite-strain plasticity framework with nonlocal damage to control mesh dependence, thickening the cortical representation

to reflect regional variation, and expanding calibration with dedicated transverse plowing tests across HU strata could elevate formulation and agreement scores to 5–6. On parameterization, formal uncertainty quantification using Bayesian calibration for cohesive and friction parameters, with propagation into outputs and decision metrics, would provide stronger grounds for higher levels. Finally, multi-laboratory replication of key tests (pullout and toggling) would establish out-of-sample performance, enabling pursuit of level 7 on measurement agreement.

Limitations in aligning manual insertion dynamics with quasi-static virtual seating were mitigated by correlating seating work rather than torque transients, but residual discrepancies will persist until dynamic and cutting phenomena are explicitly modeled. Similarly, quasi-static toggling approximates cycle-by-cycle evolution but cannot capture rate effects; the close match in degradation slopes suggests that for the selected amplitude and frequency range, the approximation is acceptable. The small-strain assumption remains a known limitation in the lowest-density bone under transverse loading; localized finite-strain updates would be the most direct remedy.

Development process and product management — gradation.

- a.
- 0: no governance
- b.
- 3: partial governance
- c.
- 5: gated plan with traceability and change management
- d.
- 7: audited governance with independent oversight

22. Limitations

The assessment is constrained by the available datasets and by the models' chosen scope. The insertion surrogate replaces a fully dynamic cutting process with quasi-static seating over one pitch; while the strong correlation for Model B supports its utility, this is not a substitute for a full rotational simulation with chip formation and rate effects. Toggling is represented as a quasi-static sequence, and although the amplitude and frequency mimic bench conditions, viscoelastic rate effects in bone are not represented. The small-strain kinematics limit fidelity when local compressive strains exceed a few percent; this primarily affects medial–lateral cut-out in the lowest-density bone, where local crushing and buckling can occur. The cortical shell is represented as a uniform layer in Model A and a uniform-thickness orthotropic layer in Model B; regional variation could be incorporated to better capture local resistance, especially in transverse plowing.

Specimen numbers per mechanism, while sufficient for the reported statistics ($n=15–30$), limit subgroup analyses (e.g., by level or sex). Although fixtures and environments were matched closely between bench and models, unmeasured compliance or friction in the experimental setup may have contributed to residual error. Furthermore, while we preserved a strict chain of custody and checksummed data, independent external audits

were not performed. We also did not conduct cross-code verification against an alternative FEA package, though canonical verification problems were executed. Finally, while we applied a structured private CRS with pre-specified sufficiency thresholds, inter-rater reliability beyond the adjudicated assessments could be strengthened by a larger panel.

23. Conclusions

We executed a mechanism-specific credibility assessment for two implicit quasi-static FEA models of thoracolumbar pedicle screw fixation across monotonic pullout, cyclic craniocaudal toggling, insertion torque–purchase, and medial–lateral shear cut-out. Model A, with thread-resolved frictional contact, reached decision-ready levels for monotonic pullout and acceptable levels for toggling and shear, but lagged in insertion-related credibility. Model B, with density mapping and cohesive interface, reached decision-ready levels for toggling and insertion and acceptable levels elsewhere. The evidence base comprised 53 thoracolumbar vertebrae with comprehensive imaging and bench records and quantitative comparisons using R2, RMSE, MAPE, and bias metrics. Coding and solver discipline and a documented development lifecycle further supported the ratings.

Our results illustrate that credibility must be argued for each mechanism that informs design decisions. Guided by the identified gaps and elevation strategies, future work will focus on interface degradation models in Model A, large-deformation updates and refined cortical representation in both models for shear, and expanded multi-laboratory validation to seek mission-ready levels. The approach and results provide a transparent basis for using these models in device R&D and for prioritizing experiments and model refinements.

24. Supplemental Material

Supplemental materials will include: (1) anonymized datasets (raw CT, segmentation masks, FE meshes, boundary condition annotations, and bench test logs) with checksums; (2) Python scripts for preprocessing, solver deck generation, and postprocessing with unit tests; (3) detailed parameter tables for cancellous and cortical bone mapping, friction and cohesive parameters, and sensitivity analysis results; (4) additional plots of model–experiment overlays by specimen for all mechanisms; and (5) traceability matrices linking requirements to verification and validation artifacts. A readme file documents environment setup and instructions to reproduce the analyses. While an external audit is not included, sufficient documentation is provided to enable independent reruns.

References

- [1] T. Silva, N. Abbott, H. Lindqvist, R. Kaur, C. Nakamura, Yield stress effects on stress range in computational models, *Med. Eng. Phys.* 144 (2022) 337–478.
- [2] B. Pettersson, K. Grigoryan, L. Ishikawa, M. Abbott, E. Silva, Cement mantle effects on cortical strain in computational models, *J. Mech. Behav. Biomed. Mater.* 90 (2015) 111–828.
- [3] F. Jankowski, S. Jankowski, Cement mantle effects on fatigue life in computational models, *Med. Eng. Phys.* 30 (2024) 43–594.
- [4] K. Kaur, D. Fontaine, K. Eriksen, Inlet profile effects on chest compression in computational models, *Med. Eng. Phys.* 123 (2014) 117–553.
- [5] R. Moreau, G. Kaur, N. Duarte, Boundary condition effects on head displacement in computational models, *Ann. Biomed. Eng.* 70 (2005) 248–685.
- [6] A. Lindqvist, E. Halvorsen, Bone density effects on fatigue life in computational models, *Ann. Biomed. Eng.* 38 (2015) 34–499.
- [7] E. Fontaine, A. Okonkwo, Yield stress effects on fatigue life in computational models, *J. Verif. Valid. Uncertain.* 53 (2015) 82–505.
- [8] C. Castellano, R. Halvorsen, A. Ueda, K. Quintero, Cement mantle effects on contact pressure in computational models, *Med. Eng. Phys.* 28 (2007) 206–843.
- [9] P. Bergman, C. Okonkwo, H. Tanaka, T. Silva, M. Tanaka, Inlet profile effects on chest compression in computational models, *J. Biomech. Eng.* 98 (2023) 137–515.
- [10] N. Ueda, G. Silva, F. Jankowski, Load path effects on hemolysis index in computational models, *J. Biomech. Eng.* 45 (2013) 108–768.
- [11] K. Rasmussen, G. Vasquez, Mesh density effects on fatigue life in computational models, *J. Verif. Valid. Uncertain.* 21 (2021) 226–434.
- [12] N. Okonkwo, T. Eriksen, F. Bergman, S. Ueda, Element type effects on shear stress in computational models, *Comput. Methods Biomech.* 69 (2017) 260–457.
- [13] K. Fontaine, C. Zetterberg, Yield stress effects on flow split in computational models, *J. Biomech. Eng.* 45 (2010) 181–868.
- [14] B. Rasmussen, E. Okonkwo, A. Vasquez, J. Jankowski, Mesh density effects on shear stress in computational models, *J. Verif. Valid. Uncertain.* 135 (2017) 29–472.
- [15] T. Yilmaz, T. Kaur, R. Nakamura, L. Eriksen, B. Abbott, Wall roughness effects on fatigue life in computational models, *Med. Eng. Phys.* 103 (2022) 127–899.
- [16] N. Fontaine, P. Zetterberg, Element type effects on outlet velocity in computational models, *Comput. Methods Biomech.* 107 (2011) 111–441.
- [17] L. Wojcik, R. Grigoryan, Gap size effects on cortical strain in computational models, *J. Verif. Valid. Uncertain.* 23 (2012) 70–506.
- [18] F. Ishikawa, C. Duarte, T. Abbott, K. Tanaka, H. Quintero, Friction effects on head displacement in computational models, *J. Biomech. Eng.* 37 (2018) 103–637.
- [19] F. Castellano, T. Lindqvist, F. Quintero, Inlet profile effects on head displacement in computational models, *J. Biomech. Eng.* 87 (2022) 400–622.
- [20] M. Ishikawa, H. Rasmussen, Gap size effects on flow split in computational models, *Comput. Fluids* 76 (2018) 337–508.

- [21] H. Ueda, D. Abbott, F. Castellano, Cement mantle effects on shear stress in computational models, *Comput. Methods Biomech.* 63 (2017) 143–543.
- [22] J. Eriksen, E. Bergman, Contact stiffness effects on shear stress in computational models, *J. Biomech. Eng.* 142 (2011) 186–828.
- [23] A. Pettersson, W. Duarte, K. Okonkwo, G. Silva, B. Tanaka, Bone density effects on outlet velocity in computational models, *J. Mech. Behav. Biomed. Mater.* 135 (2011) 127–644.
- [24] L. Castellano, N. Wojcik, D. Grigoryan, Material model effects on hemolysis index in computational models, *Comput. Fluids* 69 (2018) 260–803.
- [25] P. Castellano, K. Jankowski, N. Kaur, K. Yilmaz, W. Ishikawa, Yield stress effects on shear stress in computational models, *J. Biomech. Eng.* 142 (2020) 12–648.
- [26] A. Duarte, G. Zetterberg, Load path effects on outlet velocity in computational models, *Comput. Methods Biomech.* 97 (2021) 51–873.
- [27] M. Halvorsen, L. Castellano, Time step effects on rib deflection in computational models, *J. Verif. Valid. Uncertain.* 102 (2018) 101–417.
- [28] C. Ishikawa, W. Yilmaz, Friction effects on rib deflection in computational models, *Med. Eng. Phys.* 44 (2022) 103–745.
- [29] A. Rasmussen, W. Tanaka, M. Quintero, K. Moreau, D. Quintero, Inlet profile effects on fatigue life in computational models, *J. Mech. Behav. Biomed. Mater.* 92 (2009) 243–441.
- [30] A. Ueda, M. Ueda, E. Grigoryan, S. Ishikawa, Inlet profile effects on chest compression in computational models, *Ann. Biomed. Eng.* 48 (2008) 58–858.
- [31] F. Wojcik, J. Bergman, D. Ueda, A. Lindqvist, Contact stiffness effects on outlet velocity in computational models, *J. Mech. Behav. Biomed. Mater.* 74 (2005) 381–613.
- [32] N. Pettersson, L. Kaur, E. Quintero, J. Ueda, Boundary condition effects on peak force in computational models, *Ann. Biomed. Eng.* 142 (2009) 393–488.
- [33] B. Tanaka, B. Zetterberg, Friction effects on hemolysis index in computational models, *J. Verif. Valid. Uncertain.* 50 (2006) 84–522.
- [34] G. Tanaka, C. Wojcik, L. Ishikawa, G. Bergman, Flow rate effects on peak force in computational models, *Comput. Fluids* 36 (2016) 340–768.

Table 2: Credibility Readiness Scale (CRS) scores by factor and mechanism for Model A: Thread-Resolved Screw–Bone Contact.

Credibility factor	Level	Basis
Data pedigree - Model A - Monotonic axial pullout	6	53 T9–L5 vertebrae (69±11 y; HU 80–920) with check-summed CT and bench logs; exclusions documented; one-to-one specimen mapping (n=30).
Development process and product management - Model A - Monotonic axial pullout	5	17 requirements traced to verification/validation; 4 gated reviews (R1–R4); 132 change requests closed before scoring baseline.
Model form - Model A - Monotonic axial pullout	6	Thread-resolved augmented Lagrange contact; small-strain valid (max $\epsilon < 3\%$); Newton tol 1e6; convergence mapped for 5 mm/min ramp.
Model inputs - Model A - Monotonic axial pullout	5	$\mu = 0.28 \pm 0.06$ from tests; cancellous cap parameters from 26 cores; ± 0.04 in μ shifts peak by 7% and stiffness by 4%.
Output comparison - Model A - Monotonic axial pullout	6	Peak force $R^2 = 0.86$, RMSE=120 N (n=30); stiffness bias 6%; post-yield energy MAPE 15%.
Software quality assurance - Model A - Monotonic axial pullout	6	Abaqus 2023 HF3 frozen; 412 unit tests (86% coverage); 15 verification problems; 100% hash match across reruns.
Test conditions - Model A - Monotonic axial pullout	5	Model used 5 mm/min axial ramp, 37 ± 1 °C PBS, 3.0 mm pilot, 45 mm screw, matching bench fixtures and compliance.
Test samples - Model A - Monotonic axial pullout	6	n=30 from 53-vertebra cohort; levels T9–L5; pedicle 4.5–8.5 mm; DXA 3.7 to +0.3; allocation balanced by HU.
Data pedigree - Model A - Cyclic craniocaudal toggling	6	Toggling dataset n=18 with synchronized CT and cycle logs; environment and fixture records archived; chain-of-custody intact.
Development process and product management - Model A - Cyclic craniocaudal toggling	5	Lifecycle artifacts baselined prior to cyclic runs; deviations (2) recorded; parameter updates reviewed at R3.
Model form - Model A - Cyclic craniocaudal toggling	5	Frictional contact captures slip; quasi-static sequence with damage-free material; acceptable for cycle-averaged rocking.
Model inputs - Model A - Cyclic craniocaudal toggling	5	μ from tests; modulus mapping as in pullout; sensitivity shows ± 0.04 μ changes degradation slope by 9%.
Output comparison - Model A - Cyclic craniocaudal toggling	5	Degradation slope MAPE 18%; cycles to 0.5 mm within $\pm 1,100$ cycles across HU strata.
Software quality assurance - Model A - Cyclic craniocaudal toggling	6	Verification included quasi-static cycle energy checks; CI

Table 3: Credibility Readiness Scale (CRS) scores by factor and mechanism for Model B: Density-Mapped Damage–Plasticity with Cohesive Interface.

Credibility factor	Level	Basis
Data pedigree - Model B - Monotonic axial pullout	6	Same 53-vertebra cohort; n=30 mapped one-to-one; CT and bench logs versioned and checksummed.
Development process and product management - Model B - Monotonic axial pullout	5	Cohesive-free path for pullout documented; requirements traced; changes (5) to modulus mapping reviewed at R2.
Model form - Model B - Monotonic axial pullout	5	Anisotropic plasticity with cohesive band present; acceptable for peak capture though interface softening affects post-yield.
Model inputs - Model B - Monotonic axial pullout	5	HU-to- ρ -to-E mapping; yield scaling with $\rho^{1.50}$; cohesive parameters set but minimally active in monotonic pullout; sensitivities 5–8%.
Output comparison - Model B - Monotonic axial pullout	5	Peak force $R^2=0.79$, RMSE=160 N; stiffness bias 12%; post-yield energy MAPE 22% (n=30).
Software quality assurance - Model B - Monotonic axial pullout	6	Cohesive traction–separation verified; CI coverage 86%; input/result hashes match across platforms.
Test conditions - Model B - Monotonic axial pullout	5	Model matched 5 mm/min, 37 °C PBS, fixture compliance; pilot and screw parameters mirrored bench.
Test samples - Model B - Monotonic axial pullout	6	n=30 spanning T9–L5; HU 80–920; DXA 3.7 to +0.3; allocation balanced by density.
Data pedigree - Model B - Cyclic craniocaudal toggling	6	n=18 with complete cycle logs and CT; data integrity and environment records preserved.
Development process and product management - Model B - Cyclic craniocaudal toggling	5	Damage accumulation implementation reviewed at R3; deviations (1) documented; baseline frozen pre-validation.
Model form - Model B - Cyclic craniocaudal toggling	6	Cohesive interface with mixed-mode damage and accumulation reproduces progressive loss of attachment.
Model inputs - Model B - Cyclic craniocaudal toggling	6	Cohesive strengths $\sigma_{n0}=12$ MPa, $\tau_{t0}=5$ MPa; $G_{Ic}=0.25$ N/mm, $G_{IIc}=0.75$ N/mm; sensitivity shows ± 1 MPa shifts slope by 6%.
Output comparison - Model B - Cyclic craniocaudal toggling	6	Degradation slope MAPE 10%; cycles to 0.5 mm within ± 700 cycles; permanent slip trajectories matched strata.
Software quality assurance - Model B - Cyclic craniocaudal toggling	6	Cycle update logic unit-tested; energy consistency verified; reproducible slip increment histories.
Test conditions - Model B - Cyclic craniocaudal toggling	5	± 1.5 mm emulated quasi-statically with 50 N bias; 10,000 cycles; fixture springs replicated compliance.